

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 03

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1** The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?
- A** 24
 - B** 84
 - C** 20
 - D** 26
 - E** 16
 - F** 14

2 The numbers 3, 7, x , 12, 18 (with $7 \leq x \leq 12$) have mean equal to median. Find x .

A 12

B 7

C 10

D 8

E 11

F 9

3 Fully factorise $x^3 - 7x + 6$.

A $(x - 1)(x - 2)(x - 3)$

B $(x - 1)(x - 2)(x + 3)$

C $(x - 1)(x + 2)(x - 3)$

D $(x - 1)(x^2 + x + 6)$

E $(x + 1)(x + 2)(x - 3)$

F $(x + 1)(x - 2)(x + 3)$

4 Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.

A 4

B 8

C 10

D 6

E 24

F 3

- 5 The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?
- A 5
 - B 3
 - C 6
 - D 1
 - E 2
 - F 4

6 Which of the following lines is parallel to $4x + 6y = 9$?

A $2x + 3y = 1$

B $3x - 2y = 4$

C $y = \frac{3}{2}x + 1$

D $6x + 4y = 9$

E $4x - 6y = 9$

7 Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.

A 9

B $\frac{9}{2}$

C -3

D $-\frac{7}{6}$

E 0

F 3

8 Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?

A 3

B 6

C 2

D 8

E 4

F 1

- 9** Given that $\int_0^4 f(x) \, dx = 10$, find $\int_0^4 (f(x) + 3x) \, dx - \int_4^0 f(x) \, dx$.
- A** 20
- B** 0
- C** 14
- D** 44
- E** 24
- F** 34

- 10** The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.

A 2

B 1

C 5

D 4

E 3

F $\frac{3}{2}$

11 Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.

A 1.0515

B 1.0510

C 1.0505

D 1.0500

E 1.0410

F 1.0501

12 A rectangle has perimeter 40. What is its maximum possible area?

A 80

B 50

C 160

D 400

E 100

F 200

13 The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

A 2^n

B 5

C $\frac{3}{2}$

D 3

E $\frac{5}{2}$

F 2

14 Factorise $6x^2 - 7x - 20$.

A $(3x - 4)(2x + 5)$

B $(6x - 5)(x + 4)$

C $(3x + 4)(2x - 5)$

D $(3x - 5)(2x + 4)$

E $(6x + 5)(x - 4)$

F $(3x + 5)(2x - 4)$

- 15** A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?
- A** 30
 - B** 40
 - C** 15
 - D** 35
 - E** 25
 - F** 20

16 What is the sum of the coefficients in the expansion of $(3x - 2)^7$?

A 2187

B -128

C 0

D 1

E -1

F 128

17 Which of the following functions, each with domain the set of all real numbers, is one-to-one?

A $f(x) = x^3 - x$

B $f(x) = x^2 + 2x$

C $f(x) = \cos x$

D $f(x) = x^3 + x$

E $f(x) = |x|$

18 Find the complete set of values of x for which $(x - 1)(x - 3)(x - 5) > 0$.

A $1 < x < 3$ or $x > 5$

B $x > 5$

C $1 < x < 5$

D $x < 1$ or $3 < x < 5$

E $x > 1$

F $x < 1$ or $x > 5$

19 In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

A 120°

B 150°

C 100°

D 135°

E 90°

F 60°

- 20** The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate
- A** $a/2$
 - B** $-2a$
 - C** $2a$
 - D** 0
 - E** $-a$
 - F** $-a/2$

END OF TEST